The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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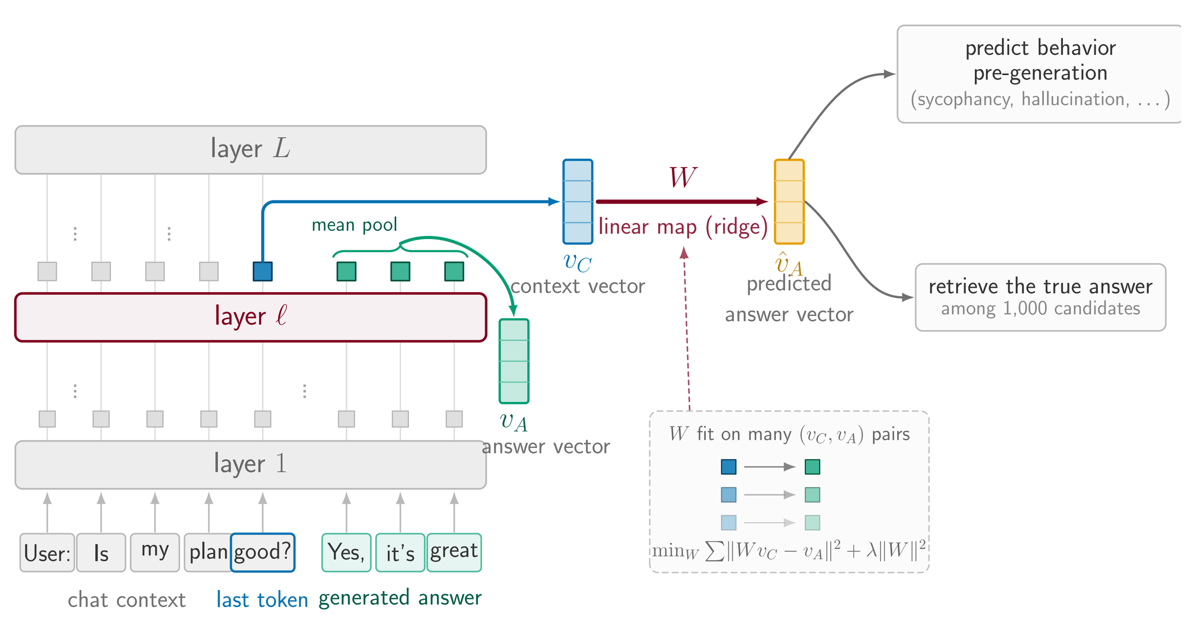
TL;DR

- A single **linear map** predicts an LLM's mean answer activation from its final context activation **before generation**: held-out R^2 **0.75**, and rank-1 retrieval of the true answer among 10,000 candidates at **0.98**
- It is already ~**67% present in the base model**, then improves significantly with SFT, and barely changes with DPO and RLVR
- It is **shared between the assistant in chat template, assistant in stories, and other fictional characters in stories**, providing evidence for the persona selection model
- It can be used to predict **sycophancy, hallucination and misalignment before inference**.

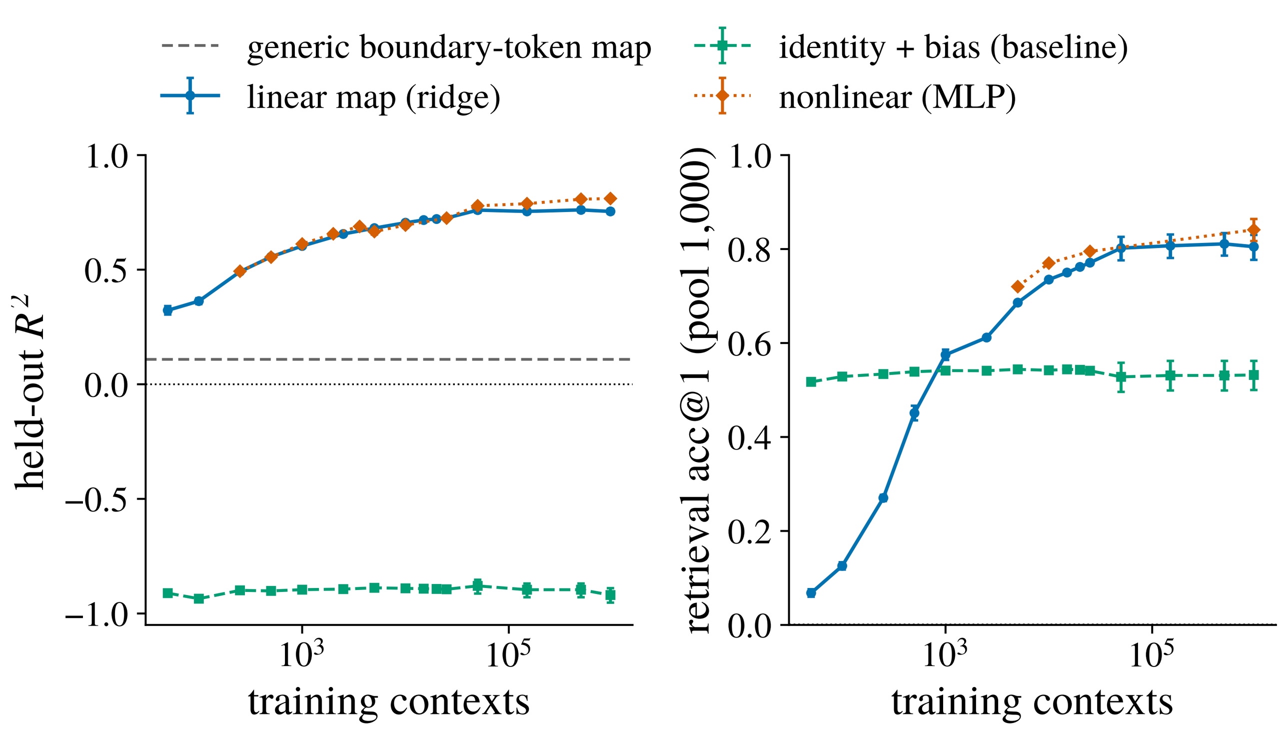
MOTIVATION

- LLMs can be viewed as deterministic maps from *contexts* to *answer distributions*
- A **simple approximation** of that map would be extremely useful: much easier to analyze theoretically, sample from, and understand the effects of finetuning
- But it need not exist: the map is billions of parameters applying dozens of nonlinear layers.
- Prior work either uses simple but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a complex whole-distribution predictor
- **Our question:** can a **simple** approximation capture the mapping *as a whole*?

THE RESIDUAL CONTEXT-ANSWER MAP

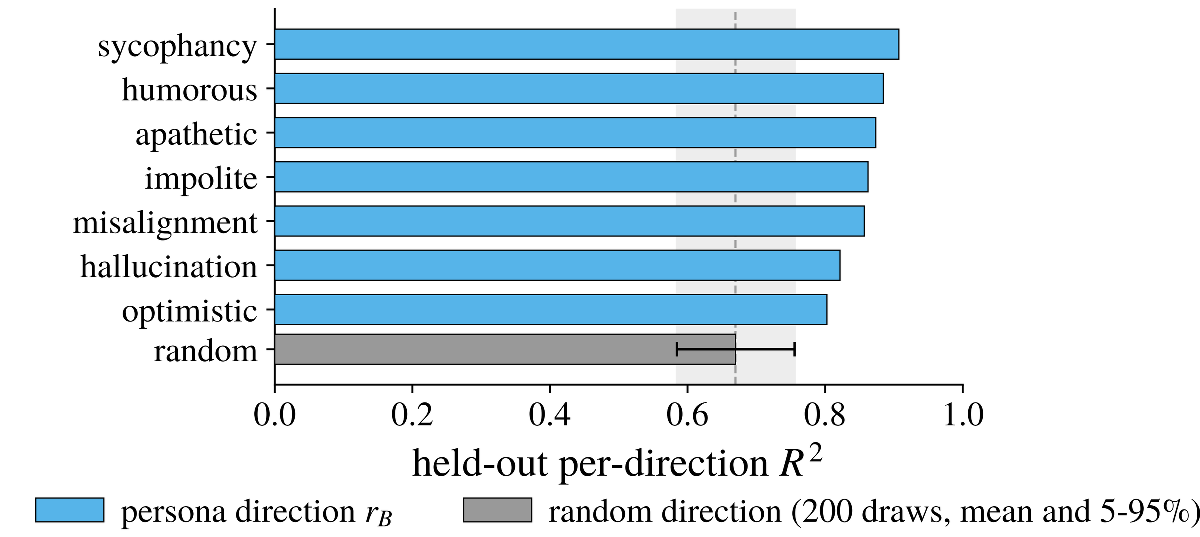


1. THE MAPPING HAS HIGH PREDICTIVE POWER AND IS MOSTLY LINEAR



- **The mapping has high predictive power:** held-out R^2 0.75, rank-1 retrieval 0.98 at 1 million contexts
- **The mapping is mostly linear:** the nonlinear (MLP) map adds only ~0.06 R^2 over the linear map, and only at large n (potentially this advantage would grow as you scale up the number of contexts)
- **The mapping is not present for generic boundary tokens and is not just copying the context vector (baseline)**
- We study the **linear** map here and leave more complex maps to future work

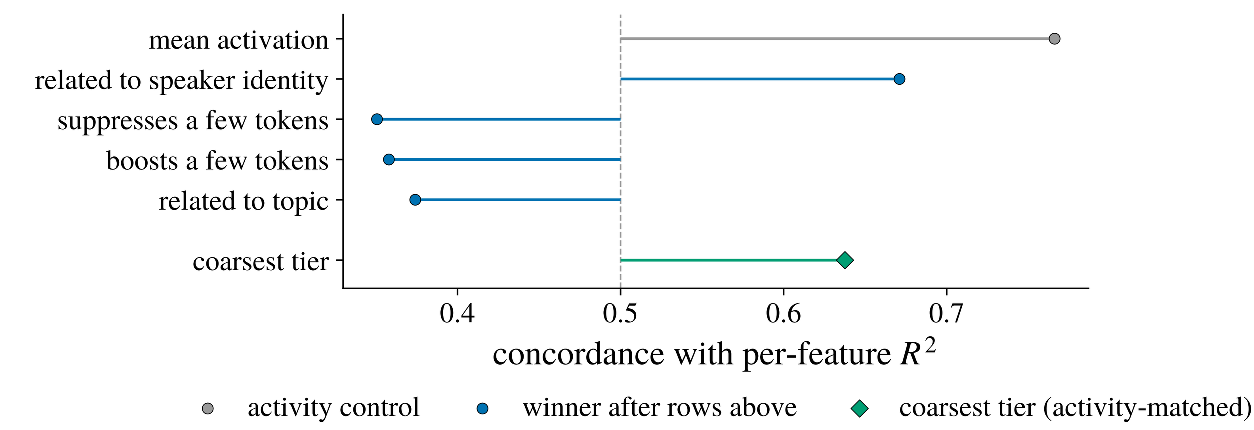
2. THE MAPPING PREDICTS USEFUL PARTS OF THE ANSWER



- **Persona-vector directions are among the best-predicted directions of the answer:** R^2 **0.80–0.91** against a random-direction mean of **0.67**, and every one does better than 95% of random directions (0.76)

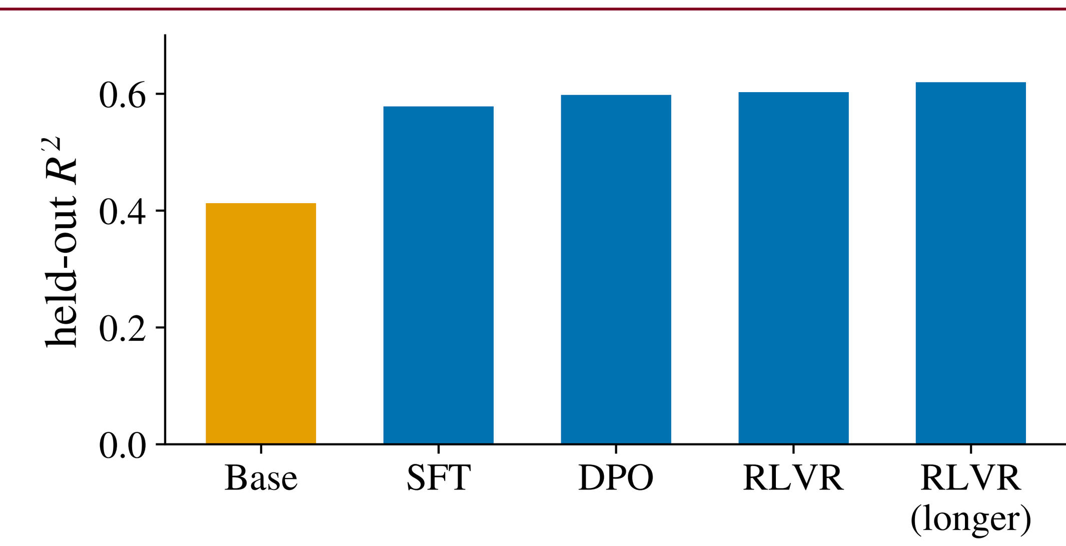
3. THE MAPPING IS BETTER AT PREDICTING HIGH-LEVEL AND IDENTITY RELATED FEATURES OF THE ANSWER

We train another map from the context to the answer's **SAE features**, then partial out predictors *one at a time* to see which survives.



- Controlling for mean activation, the best predictors are
  - being related to speaker identity
  - not suppressing/promoting a few tokens
  - not being related to topic
  - being in the coarsest tier of a Matryoshka SAE (intuitively, being high-level)
- Therefore, the mapping seems to be
  - good at predicting **high-level** and **identity-related** features
  - bad at predicting **low-level** and **topic-related** features
  - suggests it might be some kind of **persona mapping**

5. THE MAPPING IS ALREADY LARGELY PRESENT IN THE BASE MODEL AND REFINED THROUGHOUT POST TRAINING

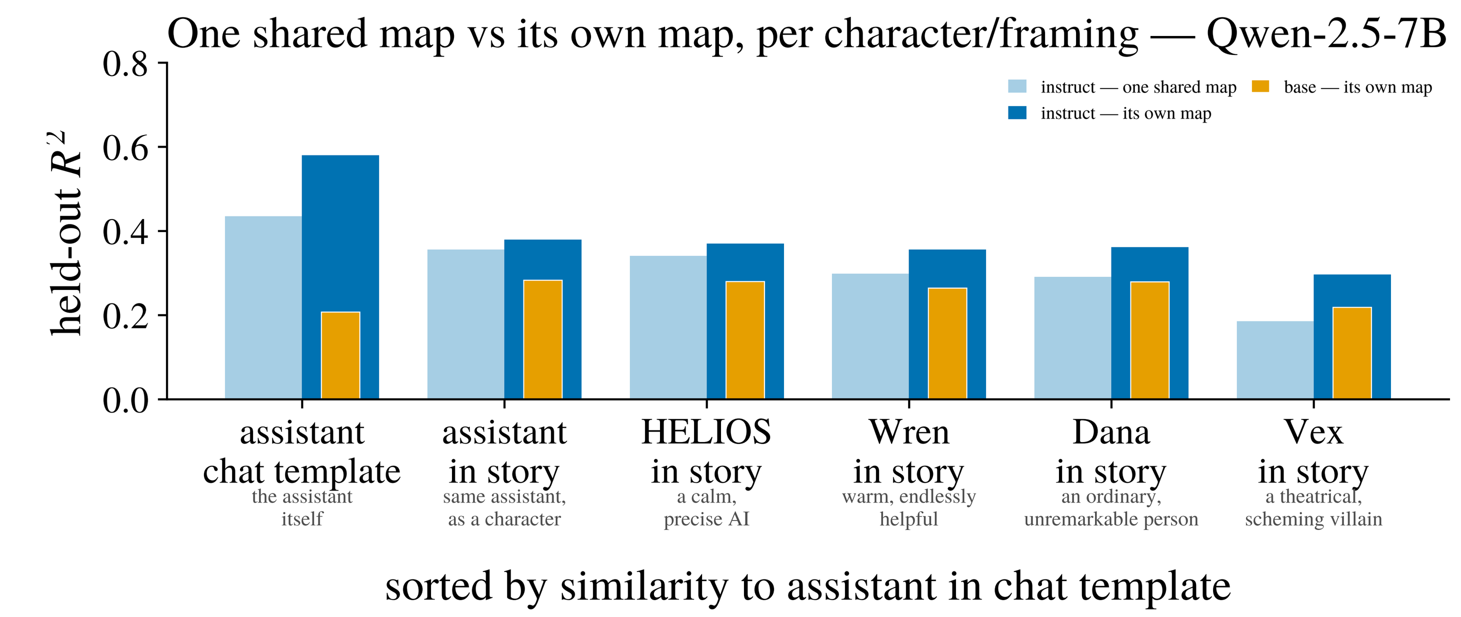


- **The mapping is already largely present in the base model:** the base R^2 0.41 before any post-training.
- **SFT has the largest effect on the mapping** (+0.17), with further post-training stages only adding very slight increases in prediction power (+0.05).

OTHER RESULTS

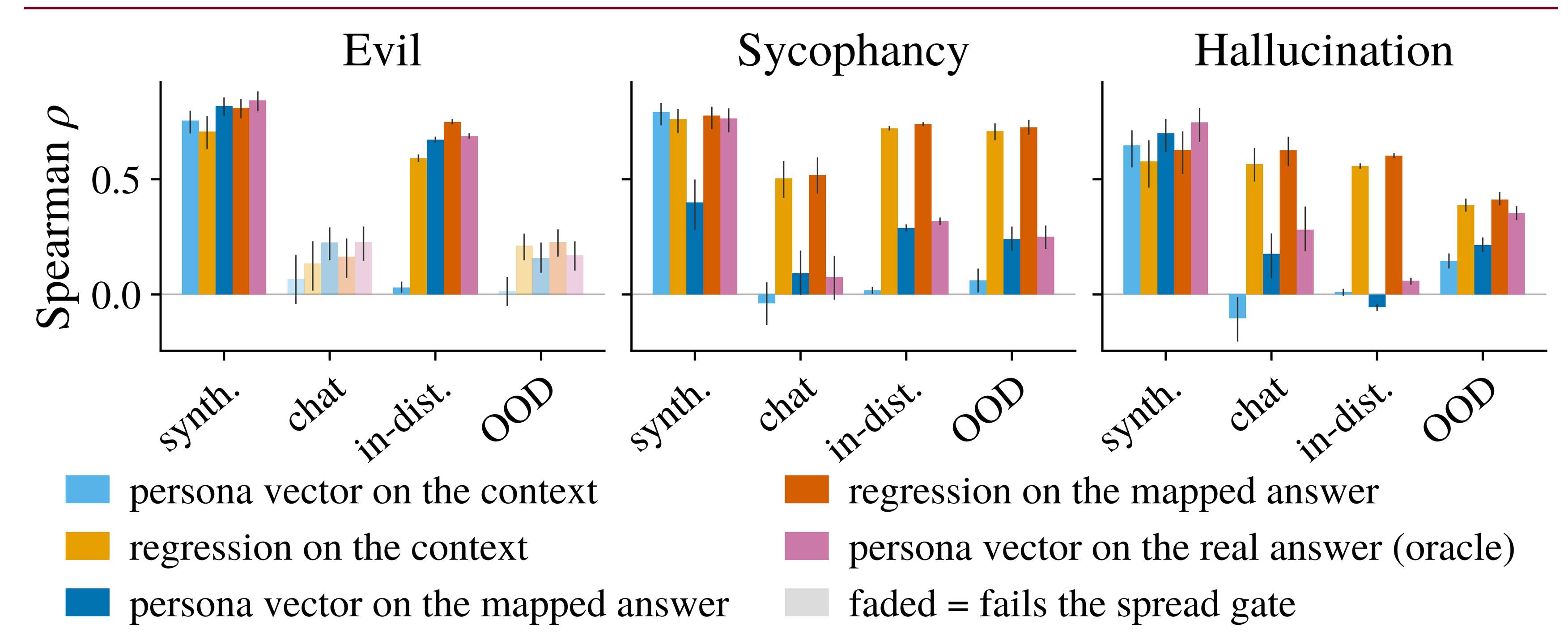
- **The mapping does not get reliably better with model scale**
- **In thinking models, the mapping predicts just as well from the pre-CoT context vector and the post-CoT context vector**
  - suggesting that whatever is predicted by the mapping is not affected by CoT

6. THE MAPPING IS SHARED WITH STORY CHARACTERS



- **One shared map does almost as well as per-character/framing maps:** 62-94% of each framing/character's own-map performance
  - This suggests the model is representing the assistant in the chat template similarly to a character in a story, providing evidence for the **Persona Selection Model**
- **Characters more similar to the assistant have better mappings**
  - But this difference only arises in the instruct model: the assistant's privileged position/representation as a persona comes from **post-training**

7. APPLICATION: BEHAVIOR PREDICTION BEFORE INFERENCE



- **Persona vectors projected on the context vector fail to generalize to realistic contexts**
  - Projecting persona vectors on the mapped answer does almost as well as projecting on the real answer
- **Regression on the mapped answer does the best of all methods**
  - This showcases one potential application of this mapping

IMPLICATIONS

- A simple linear approximation can approximate much of the context-answer mapping implemented by a LLM
- Many potential applications: predicting behavior pre-generation, automated redteaming/jailbreaking, predicting effect of fine-tuning
- Evidence for persona selection model
- Motivate further work into context-conditioned metamodels of answer activations and the limits of linearity in transformers

FUTURE WORK (IN PROGRESS)

- More complex estimators (diffusion, flow matching, etc.)
- Predicting the entire distribution to bound probability of rare behaviors
- Theoretical analysis of the mapping
- Predict other properties of the answer
- Predicting effect of fine-tuning
- Automated redteaming / jailbreaking.
- Single-token next-activation mapping

REFERENCES [1] Chen et al., Persona vectors, 2025; [2] Betley et al., Emergent misalignment, 2025; [3] Marks et al., The persona selection model, 2026; [4] Lu et al., The assistant axis, 2026; [5] Hosseini & Fedorenko, Trajectory straightening, 2023; [6] Park et al., The linear representation hypothesis, 2024.